Varnikha Varshini S

Chennai, Tamil Nadu | 7358200483 | **Varnikha04@gmail.com** |
www.linkedin.com/in/varnikhavarshini | https://leetcode.com/u/_varnikha-24/

OBJECTIVE

Passionate Computer Science undergraduate with hands-on experience in Generative AI, NLP, and Data Science, including RAG-based LLM models and machine learning models. Looking for a GenAI/Data Science opportunity to develop scalable AI solutions and unlock valuable insights from data.

EDUCATION

B.E. Computer Science Engineering (UG) **May 2026**

- Sri Venkateswara College of Engineering, Chennai *(2022–2026)*

Higher Secondary, Chennai: **2022**

- Vani Vidyalaya Senior Secondary & Junior College

Secondary School, Chennai: **2020**

- Vani Vidyalaya Senior Secondary & Junior College

CERTIFICATIONS

Machine Learning*:* Simplilearn: **September 2024**

- Completed *Introduction to Machine Learning* course covering supervised and unsupervised algorithms including Decision Trees, k-Nearest Neighbours, and Clustering.
- Implemented small ML models on sample datasets using Python and Scikit-learn, and gained practical exposure to model evaluation and accuracy improvement.

SQL: HackerRank SQL (Basic): **December 2024**

- Achieved SQL certification by demonstrating proficiency in writing and optimizing SELECT, JOIN, GROUP BY, and subqueries for data analysis.
- Practiced real-world query problems involving data extraction, manipulation, and aggregation on relational databases.

Pandas: Kaggle: **December 2024**

- Completed Pandas hands-on course covering data import/export, indexing, filtering, grouping, and handling missing values.
- Built Python scripts to clean, transform, and summarize datasets efficiently for analytical workflows.

FastAPI – The Complete Course (Beginner + Advanced): Udemy: **January 2026**

- Built high-performance REST APIs using FastAPI, including CRUD operations , authentication, and database integration.
- Developed scalable backend services with hands-on experience for real-world and AI-based applications.

PROJECTS

TTC Bus Delay Duration Predictor:

- Identified and corrected data leakage in a delay-prediction model by comparing random vs. time-based train/test splits, revealing that incident type and route drive delay duration, not weather.
- Built a regression pipeline joining 52,815 real transit delay records with historical weather via a public API, using pandas for cleaning and feature engineering.
- Achieved 23% lower prediction error than baseline (11.5 vs. 14.9 min MAE) with a random forest on log-transformed targets.
- Tech Stack: Python, Scikit-learn, pandas, Seaborn, Requests.

RAG-Powered Document Chat Application:

- Built a LLM-powered Retrieval-Augmented Generation (RAG) system using FastAPI that processes user documents and enables contextual AI conversations through a hybrid database architecture (SQLAlchemy/ChromaDB), implementing document chunking, sentence-transformers embeddings, and Google Gemini LLM integration to optimize semantic retrieval latency while ensuring user data isolation with JWT authentication.
- Tech Stack: FastAPI, Python, SQLAlchemy, SQLite, ChromaDB, Google Gemini, sentence-transformers, JWT, bcrypt.

Customer Segmentation & Retention App:

- Segmented customers into 6 behavioral groups (e.g., top-tier loyalists, dormant budget) using K-Means clustering and PCA on income, spend, and purchase-channel data.
- Deployed an interactive Streamlit app that classifies new customers in real time and generates recency-based retention recommendations per segment.
- **Tech Stack: Python, scikit-learn, pandas, Streamlit, Plotly.**

Insightly — CSV Data Analyzer:

- Developed an automated web application that parses uploaded CSVs to deliver instant data-quality reports, type-aware summary statistics, and grouped interactive charts featuring a custom sample-size reliability indicator.
- Tech Stack: Python, Streamlit, pandas, NumPy.

INTERNSHIP

Data Science — Platosys Technologies Private Limited: January 2026 – March 2026

- Engineered automated Python ETL pipelines to extract, transform, and clean unstructured data arrays for relational storage and model indexing workflows.
- Developed a conversational web application utilizing Streamlit, string-matching algorithms, and session-state persistence to automate lookup tasks.

SKILLS

Technical Skills:

- **Programming:** Python, Java, C++, SQL.
- **AI/ML:** Generative AI, RAG, LLM Frameworks (LangChain), Transformers (BERT), CNNs.
- **Frameworks:** PyTorch, TensorFlow, Scikit-learn, FastAPI

- **Tools:** Pandas, NumPy, ChromaDB, SQL
- **Web**: HTML, CSS, JavaScript

Soft Skills:

Problem-solving, Communication, Time Management, Leadership.